

2014

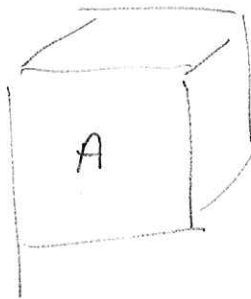
HEAT PRODUCTION

RADIOACTIVE DECAY, u , Th , K

~~INTERNAL~~ FRICTION

METAMORPHISM

MELTING (LATENT HEAT)



$$= \frac{1}{V} \frac{\Delta Q}{\Delta t} \left(\frac{W}{m^3} \right) \left(\frac{1}{m^3} \frac{J}{s} \right)$$

THERMAL CONDUCTIVITY

$$k \quad \frac{W}{m K}$$

SPECIFIC HEAT CAPACITY

$$C_p \quad \frac{J}{kg K}$$

THERMAL DIFFUSIVITY

$$\{K\} = \frac{m^2}{s} = \frac{k}{\rho C_p}$$

(2)

HEAT PRODUCTION

METAMORPHISM (PRO-GRADE)

RADIOACTIVITY, U, Th, K
 $^{238}, ^{235}U$

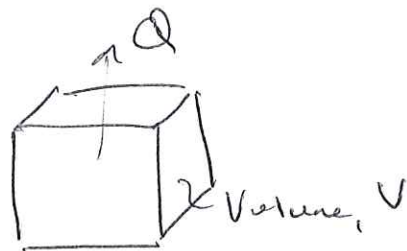
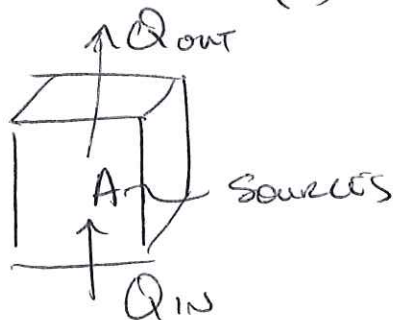
EARTHQUAKES - FRICTION

SINKS

METAMORPHISM (RETROGRADE)

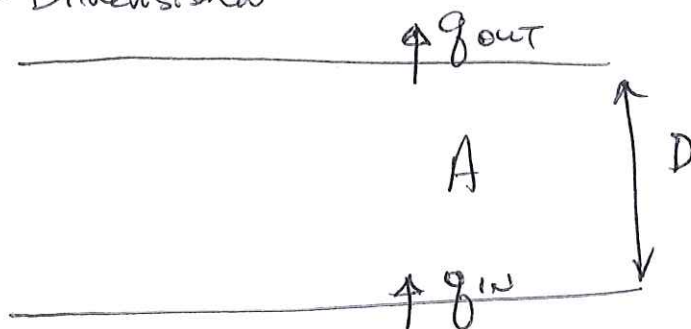
CONSERVATION OF ENERGY

• $Q = AV$ HEAT GOING OUT
IS (+)



$$\frac{\partial}{\partial t} (Q_{out} = Q_{in} + A_{sources} V)$$

1-Dimensional



$$g_{out} = g_{in} + AD$$

CONSERVATION OF
ENERGY

FOURIER'S LAW

(3)

$$\vec{q} = k \nabla T$$

DEFINITION OF HEAT FLOW

↑ GRADIENT, 1-D $\frac{\partial T}{\partial z}$

$$\frac{W}{m^2} = \frac{W}{m \cancel{K}} \frac{\cancel{K}}{m}$$

$$= \frac{W}{m^2}$$

HEAT EQUATION

$$\frac{\partial [\rho C_p T]}{\partial t} = \nabla \cdot (k \nabla T) - \nabla \cdot [\rho C_p \vec{u} T] + A$$

$$\underbrace{\rho C_p \frac{\partial T}{\partial t}}_{\text{TIME DEPENDENCE}} = \underbrace{k \nabla^2 T}_{\text{DIFFUSION}} - \underbrace{\rho C_p \nabla \cdot (\vec{u} T)}_{\text{ADVECTION}} + \underbrace{A}_{\text{SOURCES}}$$

TIME-DEPENDENCE

$$0 = k \nabla^2 T + A$$

ASSUMING NOT t -DEPENDENCE

ASSUMING $A \neq 0$

ASSUMING HOMOGENEOUS PROPERTIES

$$k(x, y, z) = k \quad A(x, y, z) = A$$

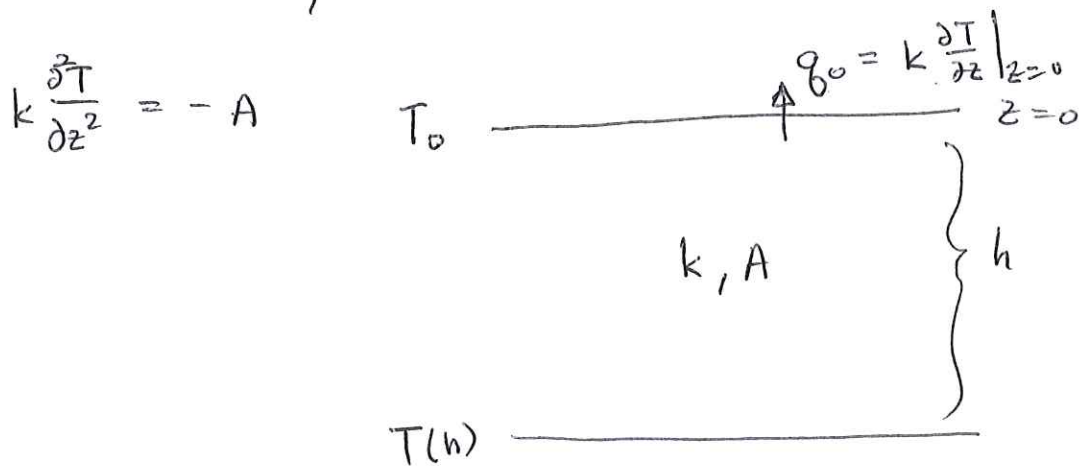
ASSUME 1-DIMENSIONAL

$$0 = k \frac{\partial^2 T}{\partial z^2} + A$$

$$k \frac{\partial^2 T}{\partial z^2} = -A$$

1-D, STEADY-STATE, CONDUCTION

④



$$\int k \frac{\partial^2 T}{\partial z^2} dz = - \int A dz$$

$$k \frac{\partial T}{\partial z} = -Az + C \quad @ z=0, \quad k \frac{\partial T}{\partial z} \Big|_{z=0} = C$$

$$q_0 = C$$

$$k \frac{\partial T}{\partial z} = -Az + q_0$$

$$\int k \frac{\partial T}{\partial z} dz = - \int (Az + q_0) dz$$

$$kT = -\frac{1}{2}Az^2 + q_0 z + C$$

$$@ z=0, T=T_0$$

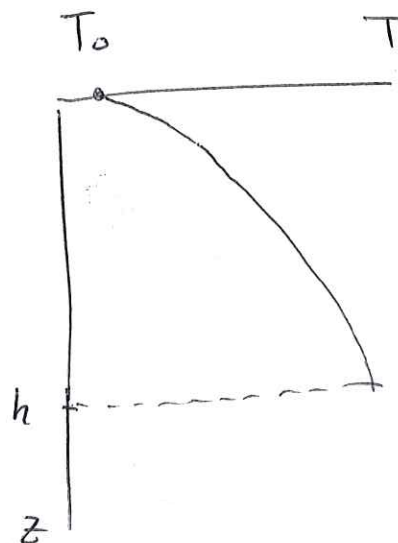
$$kT(0) = -\frac{1}{2}A(0)^2 + q_0(0) + C$$

$$T(0) = \frac{C}{k}$$

$$kT_0 = C$$

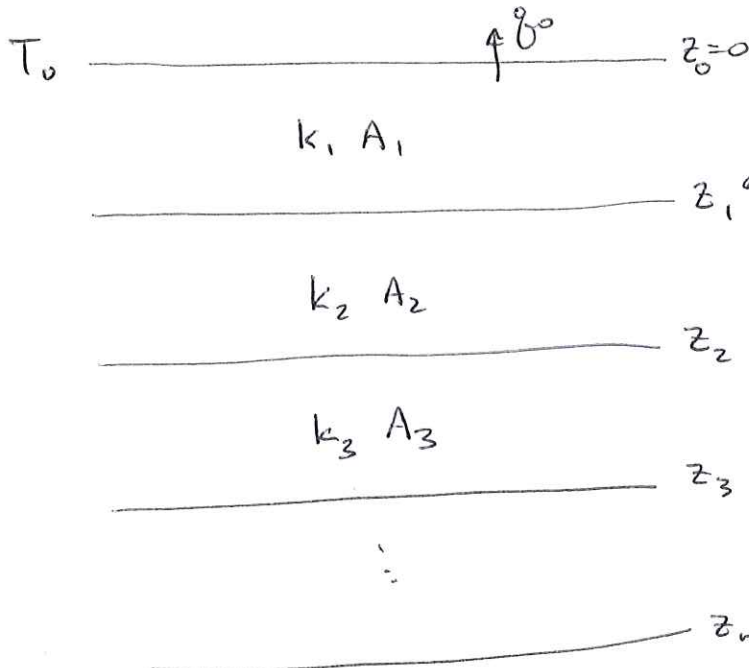
$$kT = -\frac{1}{2}Az^2 + q_0 z + kT_0$$

$$T(z) = -\frac{1}{2} \frac{A}{k} z^2 + \frac{q_0}{k} z + T_0$$



LAYERED 1-D MODEL

(5)



$$T_1 = T_0 + \frac{q_0}{k_1}(z_1 - z_0) - \frac{A_1}{2k_1}(z_1 - z_0)^2$$

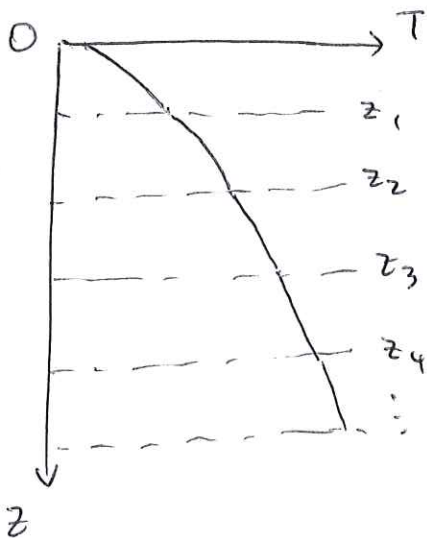
$$q_0 = q_1 + A_1(z_1 - z_0)$$

$$q_1 = q_0 - A_1(z_1 - z_0)$$

$$T_2 = T_1 + \frac{q_1}{k_2}(z_2 - z_1) - \frac{A_2}{2k_2}(z_2 - z_1)^2$$

$$q_{i+1} = q_i - A_i(z_i - z_{i-1})$$

$$T_{i+1} = T_i + \frac{q_i}{k_{i+1}}(z_{i+1} - z_i) - \frac{A_{i+1}}{2k_{i+1}}(z_{i+1} - z_i)^2$$



THERMAL CONDUCTIVITY

ROCK DENSITY, ARITHMETIC MEAN

$$\rho_{eff} = \sum_{i=1}^N \rho_i v_i = \rho_1 v_1 + \rho_2 v_2 + \dots + \rho_N v_N$$

EFFECTIVE THERMAL CONDUCTIVITY, GEOMETRIC MEAN
RANDOM CRYSTAL MIX & ORIENTATIONS

$$k_{eff} = \prod_{i=1}^N k_i^{v_i} = k_1^{v_1} k_2^{v_2} k_3^{v_3} \dots k_N^{v_N}$$

$$\sum_{i=1}^N v_i = 1$$

(6)

$$k_{\text{eff}} = \prod_{i=1}^N k_i^{v_i} = k_1^{v_1} k_2^{v_2} \dots k_N^{v_N}$$

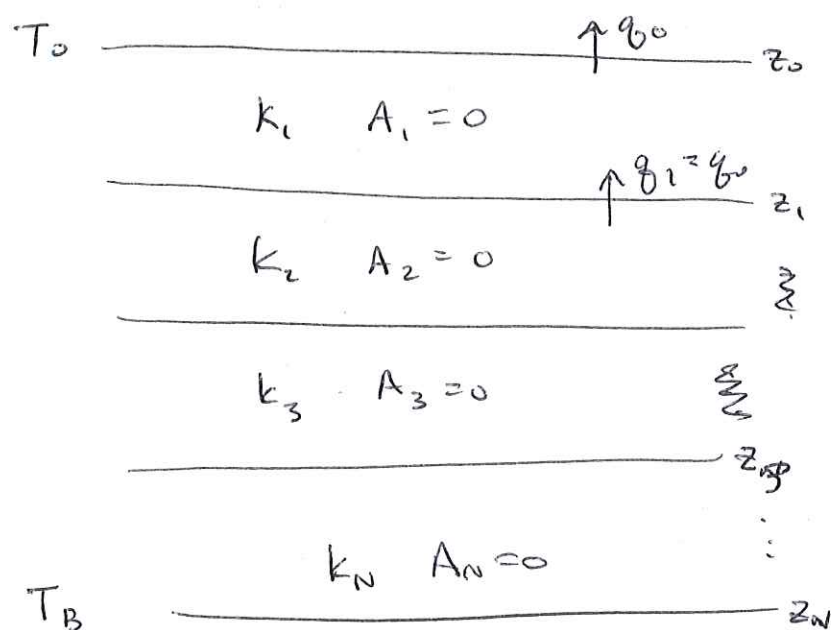
$$\log k_{\text{eff}} = \log (k_1^{v_1} k_2^{v_2} \dots k_N^{v_N})$$

$$= \log k_1^{v_1} + \log k_2^{v_2} + \dots + \log k_N^{v_N}$$

$$= v_1 \log k_1 + v_2 \log k_2 + \dots + v_N \log k_N$$

$$\log k_{\text{eff}} = \sum_{i=1}^N v_i \log k_i$$

LAYERED SOLN. NO SOURCES



$$T_1 = T_0 + \frac{q_0}{k_1} (z_1 - z_0)$$

$$T_2 = T_1 + \frac{q_0}{k_2} (z_2 - z_1)$$

$$T_3 = T_2 + \dots$$

$$T_{N-1} = T_{N-2} \dots$$

$$T_N = T_{N-1} + \frac{q_0}{k_N} (z_N - z_{N-1})$$

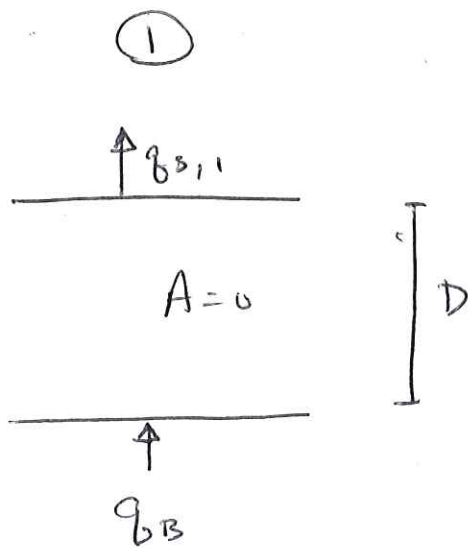
$$T_2 = T_0 + \frac{q_0}{k_1} (z_1 - z_0) + \frac{q_0}{k_2} (z_2 - z_1)$$

$$T_3 = T_0 + \frac{q_0}{k_1} (z_1 - z_0) + \frac{q_0}{k_2} (z_2 - z_1) + \frac{q_0}{k_3} (z_3 - z_2)$$

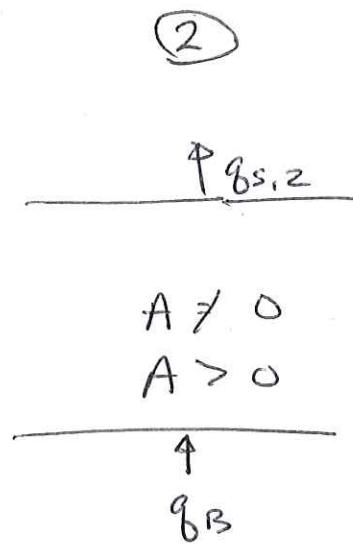
$$T_N = T_0 + q_0 \left(\frac{z_1 - z_0}{k_1} + \frac{z_2 - z_1}{k_2} + \dots + \frac{z_N - z_{N-1}}{k_N} \right)$$

$$= T_0 + q_0 \left(\frac{h_1}{k_1} + \frac{h_2}{k_2} + \dots + \frac{h_N}{k_N} \right) = T_0 + q_0 \left[\sum_{i=1}^N \frac{h_i}{k_i} \right]$$

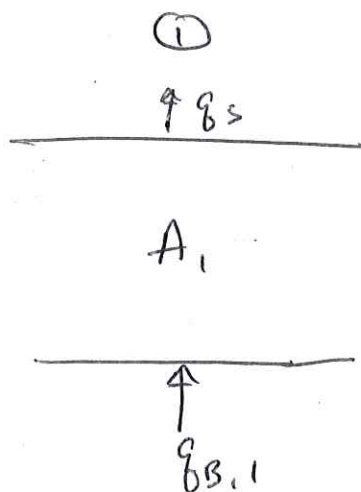
THERMAL RESISTANCE



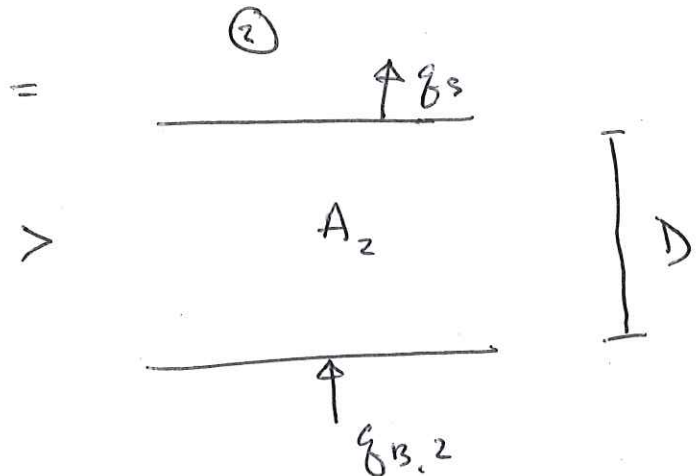
$$q_{s,1} = q_B$$



$$q_{s,2} = q_B + AD$$



$$q_s = q_{B,1} + A_1 D$$



$$q_s = q_{B,2} + A_2 D$$

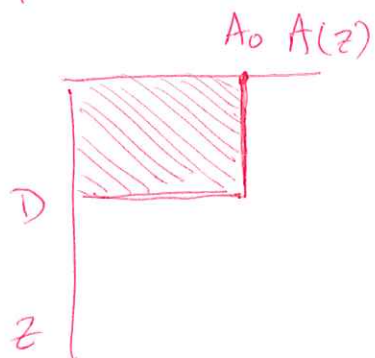
$$q_{B,1} + A_1 D = q_{B,2} + A_2 D$$

$$\cancel{A_1 D} - \cancel{A_2 D}$$

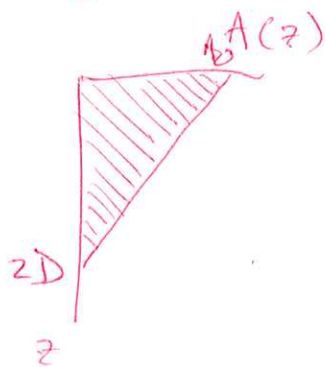
$$(A_1 - A_2) D = q_{B,2} - q_{B,1}$$

$$q_{B,2} > q_{B,1}$$

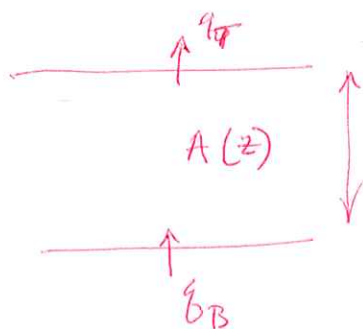
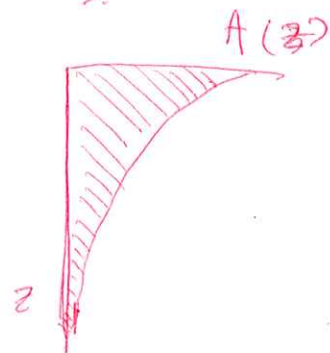
1:



2:



3:



$$g_T = \underline{A(z_B - z_T)} + g_B$$

$$g_T = \underbrace{\int A(z) dz}_{g_{RAD}} + g_B$$

$$g_{RAD} = \int_0^{\bar{z}} A(z) dz$$

AREA

$$1: A(z) = \begin{cases} A_0 & 0 \leq z \leq D \\ 0 & z > D \end{cases} \quad] \text{ ignore}$$